A Approach

Production experiment
Participants type their own utterances

↓

RSA production models
Full distribution of the fifteen possible utterances

↓

Held-out prediction
How well each model predicts each response

Controlled simulation
When does an order help the listener?

B Size Semantics × Degree of Planning

SIZE SEMANTICS

Context-fixed

big ← whole scene

big is judged against the whole scene.

Sequential context updating

big ← blue objects ← blue + N

big is judged among the blue objects.

DEGREE OF PLANNING

more planning ⟵ ⟶ less planning

Global
(fully planned)
κ = 0

big blue N

Complete utterances are compared as wholes.

Plan-guided

0 < κ < 1

value of the complete utterance

big → blue → N

The complete utterance guides each adjective.

Fully incremental

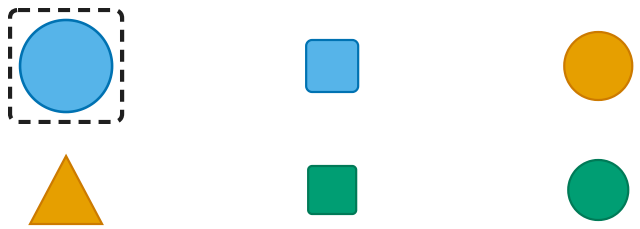
κ = 1

big → blue → N

Each adjective is compared with those available at that point.

All models assign probabilities to the same fifteen utterances.

C Referential Task



Six objects · target in the dashed frame
Here, size alone identifies the target.

POSSIBLE UTTERANCES

big | *blue* | *big blue* | *blue big* | *big blue round* | …

One to three adjectives in any order:
fifteen ordered categories of size, colour, and form

D Main Findings

DEGREE OF PLANNING

Global
fully planned

Plan-guided

Fully incremental



best-predicting model

Each tick is one speaker's estimated κ. The estimated degree of planning varies across speakers.

SIZE SEMANTICS

Fitted production
Context-fixed predicts better than updating

Controlled simulation
Updating favours size-first order only under restricted conditions

An intermediate degree of planning, varying across speakers, best predicts which utterances speakers produce.